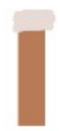
Object: Torch

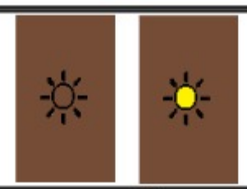
Lit	Un-Lit
	

Properties	• **Active**: lit or un-lit •
Behaviours	• Lights on fire when beside fire • Emits light when on fire

Game State Feedback	• Audio
	  • sound effect
	  • fire sounds when **Active**
	• Visual
	  • Colour
	  • turns red when lit
	  • Light
	  • emits light wen lit

Systemic Feedback	• emits light when **Active** • effects other objects when **Active**

Object: door

Light Opens	Light Close
	

Properties	• **Toggle State**: open or closed • **Light:** sun or moon lights up when conditions right • **Accessibility:** sun or Moon
Behaviours	• **Toggles** between states depending on light

Game State Feedback	• **Audio**
	  • sound effects
	  • Open/close
	  • Light react
	• **Visual**
	  • **Light**
	  • lights up when conditions met
	  • **Shape**
	  • Sun or moon

Systemic Feedback	• can be entered when **Open**

Object: flammable tile

unburnt	burning	burnt

Properties	• **Active**: burning/burnt state • **Colour**: green turns to black
Behaviours	• lights on fire when beside fire • emits light when on fire

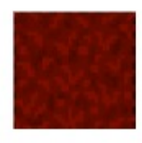
Game State Feedback	• Audio
	  • sound effects
	  • fire sounds when **Active**
	  • burnt out hiss
	• Visual
	  • colour
	  • green: unburnt
	  • red: burning
	  • Black: burnt
	  • Light
	  • Emits light when burning

Systemic Feedback	• emits light when **Active** • effects other objects when **Active**

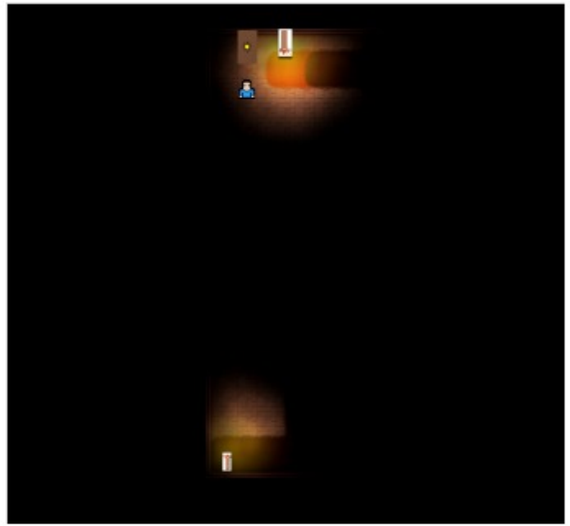
Toggle: Door

Light / Door	Light on	Light off
Sun Door		
Moon Door		

Toggle: Fire spread

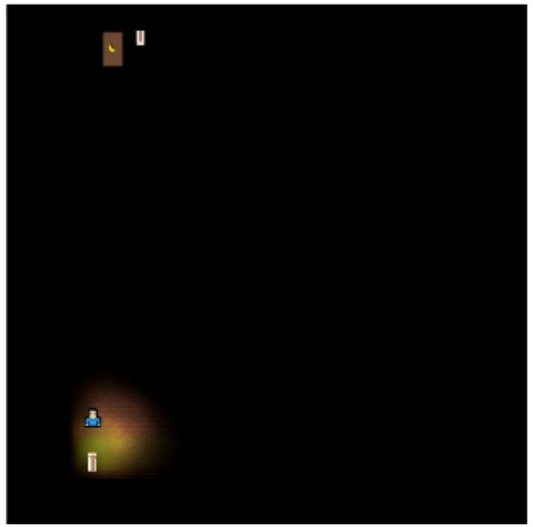
Fire / Objects	Fire beside	Fire not beside
torch		
Flammable Tile		

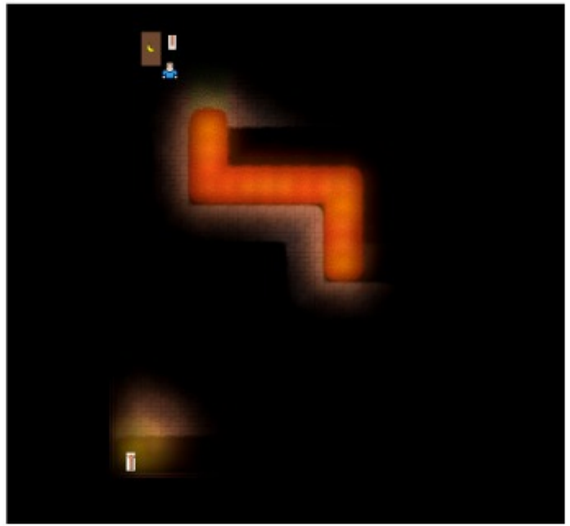
Scenario #1

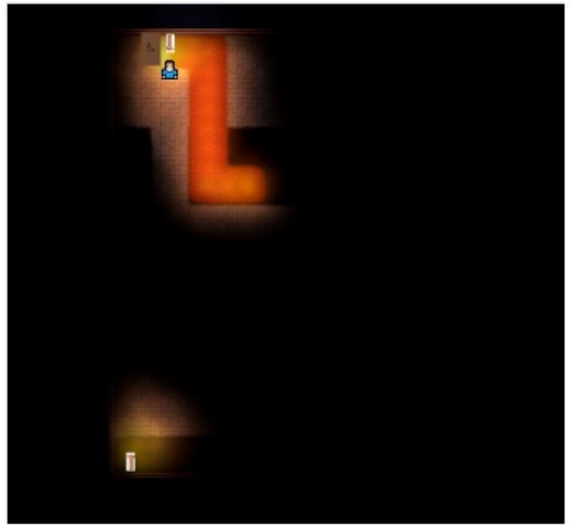


A	1. The player enters the level and sees the fire spreading 2. the door needs light to open 3. the player needs to follow the fire as it lights the way
B	1. fire burns out 2. place flammable blocks on the pit 3. cross to other side before light burns out
C	1. place flammable tile under torch
D	1. fire lights up the torch 2. door reacts to light and opens 3. level complete

Scenario #2







A	1. The player enters the level and sees the fire starting to spread 2. the door doesn't need light to open 3. the player needs to follow the fire as it lights the way
B	1. fire burns out 2. follow the flammable blocks 3. cross to other side before light burns gets to the torch
c.1	1. the flame doesn't get to the torch 2. door is open and level completed
c.2	1. flame reaches torch 2. door closes and level fails

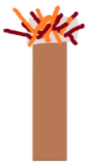
Q1. Justify Objects, Properties, Behaviours, and Relationships

For each mechanic, justify why you designed these specific objects, properties, behaviours, and relationships.

1.0 Assumptions

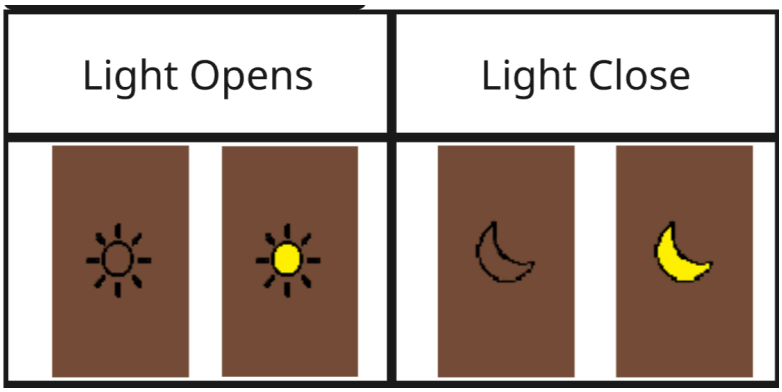
I did not list all of the mechanics that have been used in the design and scenarios. It is assumed that the fundamental actions/mechanics such as the player being able to place blocks down and having an inventory are present as a foundation for these designs.

1.1 Torch

Lit	Un-Lit
	

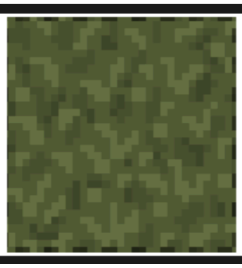
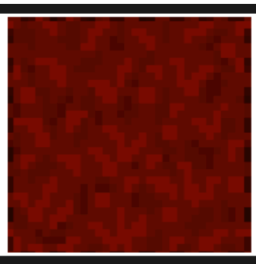
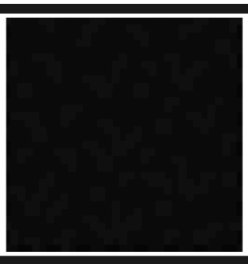
The reason the torch exists in the system is to mediate the fire converting it into light which the door then reads. The object has 2 states that can't be toggled: the lit and unlit states if it were toggleable it would be easier to break some of the puzzles. By making the object not be able to be interacted with by the player makes it so that the puzzle is environmental fire manipulation not through the torch. This object is only able to be activated once to the lit state through fire making the action irreversible forcing planning and creating a fail state for the moon door. The torch emits an environmental light within a certain range meaning that there can be multiple torches allowing layered interactions.

1.2 Door



In the system the door acts as a progression point reacting to the environmental light instead of the player. The door only reacts to the light from the torch but not the light from the burning tile because the door has a light threshold that the tile can't meet which keeps the torch as the main way to interact with the door. If the door only opened for light that would limit the amount of levels that can be created by adding a simple reverse of the object allowing for a wider array of possibilities, now light not only can open the door it can now close it to. The door continuously evaluates nearby light levels letting it dynamically change states from close to open. The symbols on each door are important to how they work. If the sun or moon weren't there the player's mental model wouldn't be supported enough. The torches placement is important to the door working if it is placed too far from the door or a nether one is too close it could react too early or not at all.

1.3 Flammable Tile

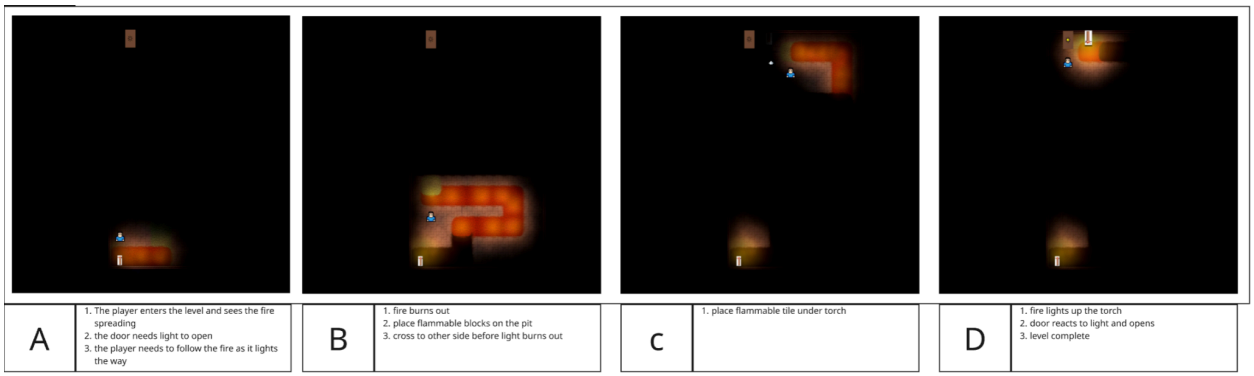
unburnt	burning	burnt
		

The flammable tile is important to the system introducing autonomous environmental change and creates time-based dependency. The three states are important for the tile to work: the unburnt allows adjacency-based spread, the burning shows that it is on fire so it can create the light and the burnt state enforces consequences and prevents infinite system cycling. The fire propagation is an automatic system that spreads at a constant pace making it predictable but uncontrollable. Since the fire can burn out it adds an urgency to the level whether it be a race against it using the fire to light the way or placing a block before it burns out Creates decision windows. The placement of the tiles in a level can change how the fire propagation works allowing the different spatial paths to create different outcomes. The player does have some control over it being given a set amount that they can place, it adds the variety of the player but also makes it a valuable resource that expands possibility space without introducing new mechanics.

Q2. Justify Emergence, Gameplay Possibility Space

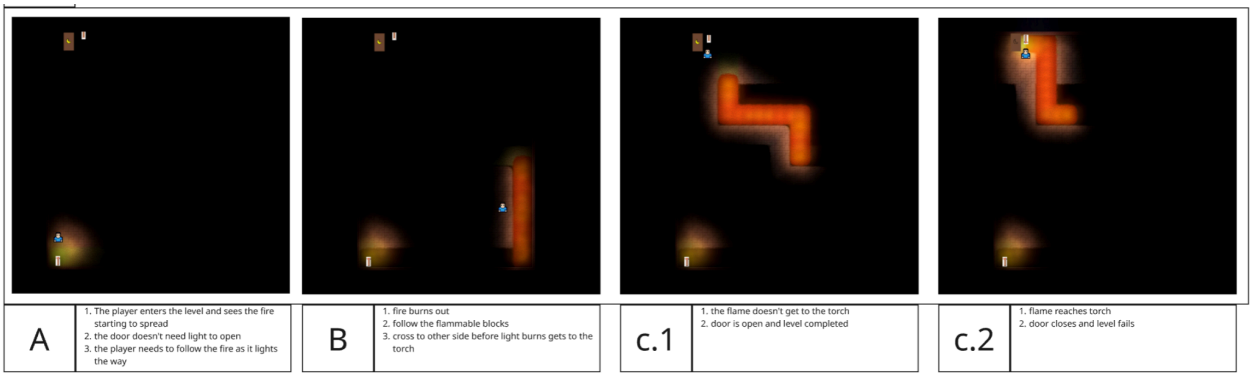
Overall, justify how you used a minimal number of objects with a sufficient number of properties to generate a wide possibility space.

Fire is important for emergence because it ties all of the objects together. Each object reacts to the fire automatically lighting on fire and emitting light or reacting to that light. The different spatial configurations change the results by changing how the fire spreads and can create a wide possibility space.



Scenario 1: Using limited flammable tiles to make a path to the torch.

In this scenario there is a starting torch that is already lit, there is a second unlit one near the sun door, there is a pit hidden by the darkness, and a limited amount of placeable tiles. The torch at the start ignites the tiles and the fire automatically propagates to adjacent tiles. The system's flow goes, fire spreads, tiles enter burning state, burning tiles emit light, Light reveals where to place the tiles, fire reaches a second torch, door reads the light, door unlocks. This isn't simple because the tiles can burn out if the limited amount of tiles aren't placed in time which could create a failed state. This scenario demonstrates emergence because the same rules can have different results depending on the layout. By moving a torch or tiles it could change the resources needed or the time to burn.



Scenario 2: Using timing and the light to make your way to the door before it closes

In this scenario there is a starting torch that is already lit, there is a second unlit one near the moon door, there is a pit hidden by the darkness, and the moon door is already open. The fire propagates automatically, tiles burn for a limited time, and the door is constantly checking for light. The system's flow goes, fire spreads, tiles burn, fire reaches the second torch, Second torch lights, light reaches the moon door, and the door closes. This isn't simple because the tiles are acting as a timer because of the burn duration so the player has to get to the other side. The player fails when the fire gets to the second torch and lights it up which closes the door. This scenario demonstrates emergence because the time pressure can be changed depending on the distance of the torches.